



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SREE SARATHI & CO

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

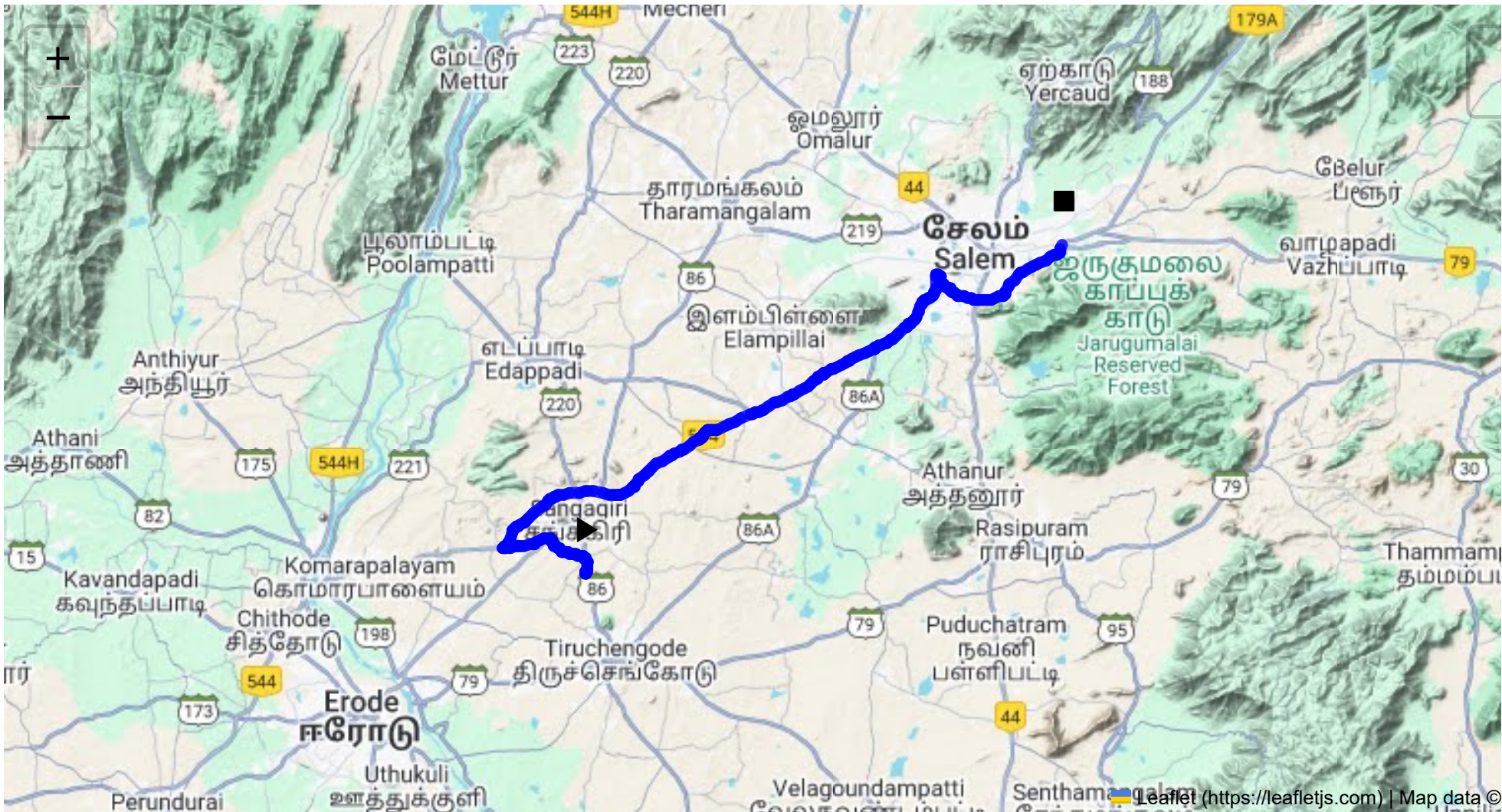
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 61.85 km
Estimated Duration: 1.2 hours
Adjusted Duration (Heavy Vehicle): 1.5 hours
Start: (11.4381, 77.8734)
End: (11.663146, 78.206268)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri to Udayapatti via specific waypoints involves a drive primarily through the state of Tamil Nadu. The journey spans approximately 61.85 kilometers and traverses various local roads, primarily along FV2C+2PH (Sankari RS - Erode Privu Rd) and Bhavani Main Road. The route features rural, semi-urban, and urban environments.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences a tropical climate. The weather can vary greatly, with hot summers, significant monsoon rains (June to September), and mild winters. Monsoon seasons can result in heavy rainfall, potentially causing waterlogging and slippery surfaces. Dense fog may occasionally occur in the mornings during winter, reducing visibility.

3. Traffic Patterns

Traffic congestion is likely to occur during peak hours, typically between 8 AM to 10 AM and 5 PM to 7 PM, especially in semi-urban and urban sections. Areas closer to Salem might experience higher congestion due to urban density. Sankari town and its surrounding areas, particularly on market days, may also witness delays.

4. Assessment of Road Quality and Infrastructure

The road quality along FV2C+2PH and Bhavani Main Road is generally decent but can vary. Potholes and uneven surfaces may be present, particularly after the monsoon. Road infrastructure includes two-lane roads for most parts, with occasional narrow stretches and potentially limited signage in rural zones.

5. Suggestions for Alternative Routes

In case of an emergency, drivers may divert through NH44, offering a more straightforward path though slightly off the primary route. Other local diversions depend on real-time conditions and accessibility.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transporting hazardous materials in Tamil Nadu requires compliance with the Motor Vehicles Act and adherence to designated routes and timings to minimize risks. Checkpoints may enforce compliance, and drivers should carry all necessary documentation.

7. Overview of Historical Incidents

The area has seen occasional accidents involving heavy vehicles, primarily due to speeding or overloaded trucks. Incidents involving hazardous materials are rare but not unheard of, emphasizing the need for cautious driving and compliance with regulations.

8. Environmental Considerations and Sensitive Areas

The route traverses rural agricultural areas; streams or water bodies can be nearby, requiring responsible handling of any spills or leaks. The driver should be aware of wildlife crossings, particularly during night hours.

9. Analysis of Communication Coverage

Mobile networks in Tamil Nadu are generally extensive, but dead zones can exist in more rural patches. Drivers should familiarize themselves with network provider coverage maps for better communication planning.

10. Estimated Emergency Response Times

In urban and semi-urban areas, emergency response times can be as quick as 15-30 minutes. In rural stretches, this can extend to 45 minutes to an hour based on the proximity of emergency services.

11. Overall Summary of Risk Assessment

This route presents moderate risk for heavy vehicles carrying hazardous materials. Weather conditions, road quality, and traffic congestion are the primary factors to consider. Ensuring compliance with local transport regulations and using technology to monitor and navigate traffic and weather conditions can significantly mitigate risks. Emergency preparedness, understanding alternative routes, and maintaining communication are essential strategies for safe transport along this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63303, 78.12713	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.45357, 77.85789	30 KM/Hr
5	Turn	Medium	11.45352, 77.85698	30 KM/Hr
6	Turn	Medium	11.46048, 77.85036	30 KM/Hr
7	Turn	Medium	11.46303, 77.84945	30 KM/Hr
8	Turn	Medium	11.46318, 77.84913	30 KM/Hr
9	Turn	High	11.45542, 77.81725	15 KM/Hr
10	Turn	High	11.45634, 77.81673	15 KM/Hr
11	Turn	Medium	11.61736, 78.10588	30 KM/Hr
12	Turn	Medium	11.61753, 78.10608	30 KM/Hr
13	Turn	Medium	11.64162, 78.12160	30 KM/Hr
14	Turn	Medium	11.64120, 78.12168	30 KM/Hr
15	Turn	Medium	11.63948, 78.12081	30 KM/Hr
16	Turn	Medium	11.65627, 78.20215	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
6	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
9	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
13	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
14	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
16	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
17	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
18	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
19	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
20	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
21	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
22	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
23	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
24	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
27	hospital	Dr.Thiru Neuro Multi Speciality	11.6623376, 78.203353	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
8	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
10	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
11	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
12	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
15	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
25	school	St.joseph's matric hr.sec school	11.6419382, 78.1709055	30 km/h	Medium
26	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.63303, 78.12713



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45357, 77.85789



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46048, 77.85036



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46318, 77.84913



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45634, 77.81673



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61736, 78.10588



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61753, 78.10608



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64162, 78.12160



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64120, 78.12168



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63948, 78.12081



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65627, 78.20215

